

NEWBERRY, SC, USA

WMO: 720596

Lat: 34.300N

Lon: 81.633W

Elev: 568

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00189

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	27.4	5.3	7.4	29.0	10.1	9.4	33.0	22.4	52.6	19.9	48.5	3.9	280	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	95.4	74.4	93.3	73.7	90.9	73.8	78.3	86.2	77.5	86.2	76.8	85.5	6.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	143.7	80.0	75.2	135.1	79.6	73.4	127.0	78.6	42.2	86.3	41.2	84.9	40.5	88.4	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.5	16.0	12.8	DB	17.8	99.3	3.5	4.0	15.3	102.1	13.2	104.4	11.2	106.7	8.7	109.5
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	43.3	47.5	53.8	63.1	71.1	78.6	80.8	79.5	74.2	62.8	52.1	47.4	(6)
	DBStd	15.00	9.53	9.28	9.03	7.38	6.27	4.15	3.47	3.67	5.42	6.97	7.51	8.57	(7)
	HDD50	693	255	147	65	4	0	0	0	0	0	3	61	158	(8)
	HDD65	2745	674	493	363	123	25	0	0	0	5	123	391	548	(9)
	CDD50	5408	46	77	182	396	654	857	955	914	727	399	124	77	(10)
	CDD65	1989	1	4	15	65	214	407	490	449	283	54	4	3	(11)
	CDH74	18422	4	20	140	652	1822	3984	4928	4131	2265	456	17	3	(12)
	CDH80	7837	0	0	17	140	641	1840	2379	1848	899	73	0	0	(13)
(14) Wind	WSAvg	5.2	6.1	6.5	6.3	6.2	5.0	4.6	4.5	4.1	4.4	5.1	4.9	5.4	(14)
(15) Precipitation	PrecAvg	45.3	4.6	3.9	4.6	3.0	3.2	4.2	3.8	4.4	4.0	3.2	3.0	3.3	(15)
	PrecMax	58.8	14.2	7.4	11.7	7.0	6.2	10.9	8.3	17.0	8.9	9.6	10.4	8.4	(16)
	PrecMin	34.1	0.6	0.9	0.7	0.3	1.1	1.1	0.6	0.4	0.2	0.0	0.6	0.4	(17)
	PrecStd	6.9	2.3	1.8	2.2	1.8	1.5	2.4	2.0	3.0	2.1	2.6	1.9	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.6	77.2	82.0	87.7	94.6	97.4	99.2	97.1	93.5	85.8	76.8	72.9	(19)
		MCWB	63.0	63.2	62.1	65.5	70.5	73.7	74.7	72.8	71.9	69.1	64.2	65.0	(20)
	2%	DB	67.7	72.0	78.8	83.7	90.0	95.1	96.7	94.6	90.6	81.8	72.2	69.9	(21)
		MCWB	59.9	60.2	61.5	65.4	69.4	74.4	74.4	74.0	71.6	67.2	61.1	64.0	(22)
	5%	DB	63.4	67.7	73.1	81.3	86.5	91.2	93.5	91.0	88.4	79.2	69.6	64.4	(23)
		MCWB	56.8	59.0	58.9	64.7	69.3	73.7	74.4	74.6	71.8	64.9	60.1	61.2	(24)
	10%	DB	59.2	63.1	70.1	78.6	82.4	89.9	90.8	89.8	86.0	76.6	65.5	62.7	(25)
		MCWB	53.4	56.3	58.1	63.3	69.1	73.5	74.8	74.4	71.4	65.1	58.3	58.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.2	66.7	66.0	72.0	74.7	78.3	79.1	80.0	77.5	75.2	68.1	68.7	(27)
		MCDB	70.4	73.4	74.0	78.3	82.9	88.8	86.6	85.5	84.5	80.4	70.8	70.8	(28)
	2%	WB	62.7	63.9	64.2	69.7	73.1	77.2	78.2	78.5	75.7	73.1	65.6	65.4	(29)
		MCDB	65.3	67.3	72.1	74.9	81.9	87.9	87.0	85.2	83.4	75.7	70.3	67.8	(30)
	5%	WB	59.1	61.4	62.6	67.9	72.0	76.2	77.4	77.7	74.4	70.4	63.2	62.8	(31)
		MCDB	62.1	66.0	69.6	75.8	80.6	86.7	87.2	84.8	81.9	74.4	65.5	64.3	(32)
	10%	WB	54.4	57.6	60.6	65.7	71.0	75.1	76.6	76.9	73.4	67.5	60.6	59.4	(33)
		MCDB	57.4	62.5	67.2	75.9	78.9	85.1	86.1	84.3	80.6	73.3	64.4	63.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	19.6	22.0	23.7	20.7	20.4	20.2	19.0	19.5	22.3	22.0	18.1	(35)
		MCDBR	23.9	22.8	26.3	25.9	24.5	23.2	23.5	22.0	23.1	25.5	25.0	20.0	(36)
	5% WB	MCWBR	16.1	13.3	12.5	10.5	8.4	6.7	6.1	6.2	7.2	11.4	15.0	13.8	(37)
		MCWBR	20.0	18.3	21.3	20.6	18.4	19.9	19.4	17.5	18.6	16.9	17.3	16.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.318	0.327	0.366	0.406	0.466	0.495	0.543	0.525	0.438	0.396	0.345	0.320	(40)
		taud	2.497	2.470	2.391	2.267	2.145	2.107	1.994	2.034	2.258	2.347	2.450	2.525	(41)
	Ebn at Noon		283	292	288	279	263	254	241	243	261	264	270	276	(42)
		Edh at Noon	27	30	36	43	49	50	56	53	41	34	28	25	(43)
(44) All-Sky Solar Radiation	RadAvg		865	1064	1406	1765	1942	2040	1921	1758	1511	1256	954	747	(44)
	RadStd		46	117	102	134	155	146	127	85	132	151	78	72	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+317	+216

Nomenclature: See separate page